

Arka Mukherjee

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- Research Focus: LLM reasoning, multimodality, AI agents, and LLM inference with a human-computer and agent-computer interaction focus

Education

Kalinga Institute of Industrial Technology (KIIT) Bengaluru, India
Bachelor of Technology in Computer Science; Cumulative GPA: 9.74/10.0 July 2023 - Apr 2027 (Expected)
Relevant Courses: Agentic AI, Probability & Statistics, Data Analytics, Machine Learning, AI, Algorithms, Data Mining, Human-Computer Interaction (Coursera)

Experience

AMD - AGI Team Bengaluru, India
Incoming Research Intern Incoming
LLM Inference: Working on LLM inference.
University of Southern California - Dr. Maja Mataric Bengaluru, India
IUSSTF-Viterbi Summer Research Intern May - August 2026
Long Video Comprehension: Built LazySloth, a fast video understanding method that improved retrieval efficiency by up to 8.3x over existing pipelines such as VideoLucy and WorldMM, while retaining task accuracy. Contributed to lab infrastructure on LLM inference. (Submitted to AAAI 2027)
Carnegie Mellon University - Dr. Min Xu Remote
Research Intern January - May 2026
Reinforcement Learning: Applied GRPO, DPO, and PPO to study RL generalization to codon translation, a low-sampling decoding task (64-token action space). Improved VRAM efficiency of the code to iterate GRPO on 800k+ datapoints.
IIT Bhubaneswar - Dr. Shreya Ghosh Bengaluru, India
Undergraduate Research Fellow (Funded) December 2024 - Present
Multimodal STEM reasoning: Developed mmJEE-Eval, a multimodal reasoning benchmark of 1,460 STEM problems to combat saturation. Evaluated 17 VLMs and discovered metacognitive barriers: VLMs detect 53% of errors but correct only 3.5%. (IJCNLP-AAACL 2025)
Multimodal Cultural Competence: Created the first systematic evaluation framework for VLM cultural competence via story generation on 5 VLMs with a novel evaluation metric. (Oral @ ICCV 2025 ASI workshop)
Integrated Capability Evaluation (ICE): Proposed ICE, a new evaluation paradigm unlike benchmarks and leaderboards that models real-world tasks. Engineered CoTDL-mini-SWE-agent, a novel coding harness to improve frontier agentic terminal coding performance. (Submitted to NeurIPS 2026)
IIT Ropar - Dr. Sudarshan Iyengar Ropar, India
Summer Research Fellow (IASc-INSNA-NASI) May 2025 - July 2025
VLM Benchmarks: Developed EduVLM-Bench for STEM prerequisite detection and evaluated OSS LLMs. The top model, Gemma3 27B, achieved 38.5% accuracy.

Publications

- 1. Mukherjee, A., Roy, S., Trivedi, K., & Ghosh, S. (2026). GeoAgent: Evaluating VLM Geolocalization Through Embodied Navigation (EMNLP 2026 Findings, 14.3% acceptance rate)
- 2. Mukherjee, A., & Ghosh, S. (2025). mmJEE-Eval: A Bilingual Multimodal Benchmark for Evaluating Scientific Reasoning in Vision-Language Models (IJCNLP-AAACL 2025 Findings, 12% acceptance rate)
- 3. Mukherjee, A., & Ghosh, S. (2025). Toward Socially Aware Vision-Language Models: Evaluating Cultural Competence Through Multimodal Story Generation (ICCV 2025 ASI workshop - Oral)
- 4. Mukherjee, A., & Ghosh, S. (2026). ICE: Heterogeneously Composing Tasks to Evaluate Closed-Loop Coordination in LLMs (Submitted to NeurIPS 2026)

Projects

- Automated VLM reasoning dataset generator: Curated 10k samples for SFTing VLMs. Our method achieved the best VMCBench results, beating Walton Cold Start and MM MathInstruct. (Ranked #6 in NeurIPS DCVLR challenge)

Achievements/Extracurriculars

- Awards: Singapore AI Safety Hub (SASH) Frontier AI Security Training (FAST) fellow (fully funded, 2.8% acceptance rate), KIIT Merit Scholar (Dean's list) for 6 semesters, Amgen Scholars Summer Research Program (declined), Aalto Science Institute (ASci) Summer Research Program in Finland (declined)
- Outreach: Published articles (10M+ reads) and hardware reviews in collaboration with Nvidia, AMD, and ASUS.
- Nvidia GPU Sponsorship: Received RTX 5090, 5080, and 5070 GPUs from Nvidia India for AI research.
- Travel Awards & Grants: D&I Subsidy award at IJCNLP-AAACL'25 (\$750), AI Grants India (API keys and credits)

Technologies and Skills

- Languages: Python (advanced), Java, C, SQL
- ML/NLP Frameworks: PyTorch, HF Transformers, Datasets & TRL, Unsloth, LM Studio, Docker, Agentic AI (LangChain, Playwright, mini-SWE-agent)